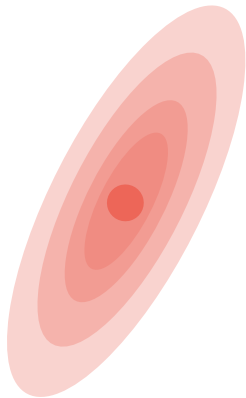


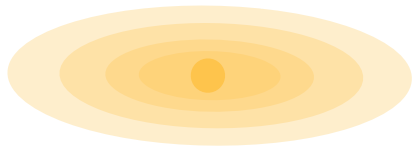
ρ_1



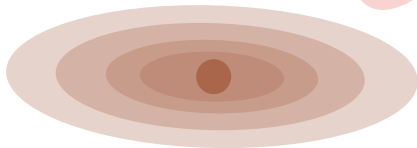
ρ_2



ρ_3



ρ_4

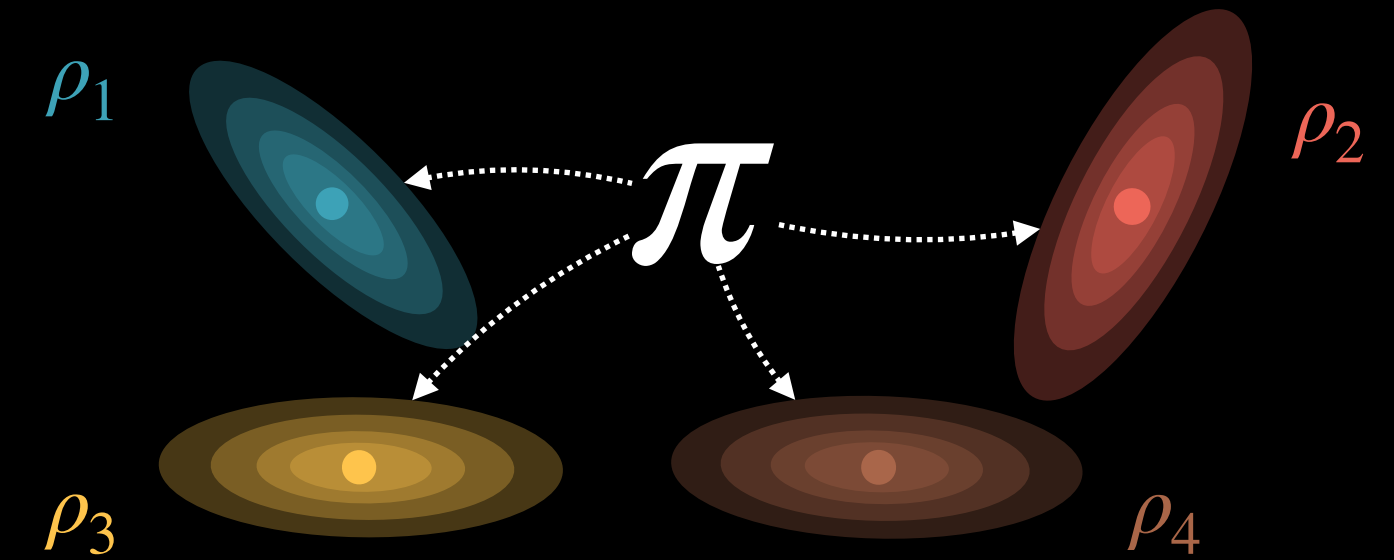


Multi-Marginal OT: Formulation

- **Problem:**

General measures:
$$\min_{\pi \in \Pi(\rho_1, \dots, \rho_S)} \int_{X_1 \times \dots \times X_S} c(x_1, \dots, x_S) d\pi(x_1, \dots, x_S)$$

Discrete measures:
$$\min_{\pi \in \Pi(a_1, \dots, a_S)} \langle C, \pi \rangle \triangleq \sum_s \sum_{i_s=1}^S C_{i_1, \dots, i_S} \pi_{i_1, \dots, i_S}, \quad C_{i_1, \dots, i_S} = c(x_{i_1}, \dots, x_{i_S})$$



- **Set of Feasible Multi-Couplings:**

General measures:
$$\Pi(\rho_s)_{s=1}^S = \left\{ \pi \in \mathcal{M}_+^1(X_1 \times \dots \times X_S) : (\text{Proj}_{X_s})_{\#} \pi = \rho_s, \forall s \in \llbracket S \rrbracket \right\}$$

Discrete measures:
$$\Pi(\mathbf{a}_s)_{s=1}^S = \left\{ \pi \in \mathbb{R}_+^{n_1 \times \dots \times n_S} : \forall s \in \llbracket S \rrbracket, \forall i_s \in \llbracket n_s \rrbracket; \sum_{i_1, \dots, i_{s-1}, i_{s+1}, \dots, i_S} \pi_{i_1, \dots, i_S} = (\mathbf{a}_s)_{i_s} \right\}$$

- **Special cases:**

- recovers standard OT (Kantorovich) when $n = 2$
- barycenter problem when taking of the marginals to be the barycenter, fix the other ones, optimize

Pop-Quiz

Wasserstein Barycenters

- Extends means/centroids to distributions
- Like Frechet mean, but for distributions
- Barycenter: $\nu^\star = \arg \min_{\nu} \sum_{i=1}^K \lambda_i W_p^p(\mu_i, \nu)$

Show how to write the barycenter problem as multi-marginal OT one